JAZTON TEOH WEE KUAN

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EDUCATION

Singapore Institute of Technology
Sep 2025 - Expected 2028

Bachelor of Engineering with Honours in Computer Engineering

- Developing practical knowledge in embedded systems, programming, electronics, sensors and hardware-software integration.
- Applying engineering and software concepts through hands-on projects in IoT systems, FPGA development and object-oriented programming.

Nanyang Polytechnic | School of Engineering
Apr 2020 - Mar 2023

Diploma in Electronic & Computer Engineering

- Director's List, Year 3 Semester 2.

WORK EXPERIENCE

Operations Team Member Intern | STYL Solutions Pte Ltd
Mar 2022 - Aug 2022

- Deployed and configured Smart Buddy cashless payment kiosks across schools, including on-site installation, system setup and operational checks.
- Provided technical and customer support for payment and smartwatch issues, troubleshooting faults and responding to service enquiries.
- Supported Automatic Tray Return System deployments at Market Street at CapitaSpring and Margaret Drive Hawker Centre as the on-site liaison for remote IT technicians.
- Installed system software, registered and paired machines, adjusted settings and performed functional testing to verify successful deployment.
- Configured Halal and Non-Halal colour detection and 4G connectivity, and carried out on-site troubleshooting with remote IT support.

PROJECTS

IoT First Aid Kit Monitoring System

Singapore Institute of Technology

- Developed an IoT monitoring system using an ESP32-C3 microcontroller and magnetic reed switch to detect and record first-aid kit openings.
- Implemented Wi-Fi communication with a Raspberry Pi-based server to transmit opening events and battery information for monitoring.
- Developed a web-based interface for viewing recorded activity and system information.
- Integrated battery-voltage monitoring, power-saving behaviour and a rechargeable LiPo battery, then designed a compact enclosure for the electronics and wiring.

PROJECTS

FPGA Based Smart Traffic Light Controller

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- Designed an FPGA-based intersection controller using a finite-state machine, combinational logic and sequential logic.
- Developed a four-state traffic sequence with Gray-code state encoding to coordinate main-road and side-road signals based on vehicle detection.
- Integrated IR vehicle detection, timed transitions, 7-segment countdown and state displays, pedestrian controls, LEDs and a buzzer.
- Validated the digital logic through simulation and implemented the completed design on a Basys 3 FPGA board.

C++ Driverless Shuttle Route Planner

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- Developed a C++ route-planning application that matches passenger requests with driverless electric shuttles by destination, requested arrival time and seating capacity.
- Applied abstraction, encapsulation, inheritance and polymorphism to structure the application.
- Implemented scheduling strategies that prioritise either minimum shuttle dispatches or passenger arrival time while accounting for capacity, group size and arrival constraints.
- Imported shuttle and passenger data, managed records in memory and exported generated route plans to external files.

TECHNICAL SKILLS

Programming: C/C++, Python, SQL, Object-Oriented Programming

Embedded Systems and IoT: ESP32, Arduino, Raspberry Pi, microcontrollers, embedded systems, Internet of Things

Hardware and Electronics: Sensors, electronics, hardware-software integration

Software and Tools: GitHub, Microsoft Excel

Additional: Technical troubleshooting, technical documentation

LEADERSHIP AND COMMUNITY ENGAGEMENT

Silver Generation Ambassador | Agency for Integrated Care — Jul 2026 - Present

- Conduct home visits and engage seniors to understand their needs, well-being and circumstances while building rapport through meaningful conversations.
- Share information on relevant government schemes, healthcare resources, community programmes and support services.
- Identify seniors who may benefit from additional assistance and connect them with appropriate community and care resources.
- Develop communication, active listening and interpersonal skills through regular engagement with seniors from diverse backgrounds.

Student Ambassador | Nanyang Polytechnic — Jan 2023

- Represented Nanyang Polytechnic during Open House, engaging prospective students and visitors while supporting event activities.